Hitesh Rasineni

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ABOUT ME

CS undergraduate working at the intersection of Machine Learning, statistics, and High Energy Physics, with experience in density estimation (normalizing flows, likelihood-ratio methods) for collider data and in building cloud-native ML systems — experiment-orchestration platforms (FastAPI, Kubernetes, MLflow, Prometheus/Grafana) and query-routing RAG assistants over experiment tracking. Focused on advancing density estimation techniques and applying them to Beyond the Standard Model (BSM) physics, backed by production-grade MLOps.

EDUCATION

VIT-AP University, Amaravati, Andhra Pradesh, India

September 2023 — September 2027

B.Tech, CSE (AI & ML)

CGPA: 8.05 / 10

Sree Vidyanikethan International School, Tirupati, India

2023

CBSE (10+2)

Grade: 91.6 / 100

SELECTED PROJECTS

The Unlikely Lab — Personal Research Publication Platform

the-unlikely-lab.github.io/the-unlikely-lab

Code: github.com/The-Unlikely-Lab/the-unlikely-lab

- Designed and deployed a full research-publication platform that hosts complete, typeset HTML versions of arXiv-style physics papers (sections, numbered equations, figures, tables, bibliography) alongside an MDX-driven content system for research notes and project pages.
- Built an automated LaTeX-to-HTML paper build pipeline with Pandoc, including MathML/KaTeX rendering, figure/table validation reports, and per-paper static generation; the site is a statically exported Next.js (React, TypeScript, Tailwind CSS) application deployed via GitHub Actions CI/CD to GitHub Pages.

Cloud-Native ML Training Platform

FastAPI, Docker, Kubernetes, Next.js, PyTorch, MLflow

Code: github.com/HiteshRasineni/Cloud-Native-ML-Training-Platform-for-Scientific-Data

- Built a cloud-native ML experiment-orchestration platform for HEP data with a full vertical slice: a FastAPI backend (PostgreSQL/Alembic + Redis) validates specs, resolves dataset references against a registry, and enqueues jobs; the scheduler runs a QUEUED → SCHEDULING → RUNNING → COMPLETED/FAILED lifecycle (retries, heartbeats, failure classification) via a pluggable executor — LocalDockerExecutor and a KubernetesExecutor (indexed batch/v1 Jobs, per-job ConfigMaps, kind/minikube manifests) — launching workers that report per-epoch metrics to MLflow + Redis; the Next.js/TypeScript frontend submits experiments and renders loss curves through a backend MLflow proxy.
- Separated storage so PostgreSQL owns lifecycle/scheduling state, the dataset registry and artifact index, MLflow (SQLite + MinIO artifacts) owns params, metrics and model artifacts, and MinIO owns datasets, checkpoints and logs; a workload registry runs a from-scratch RealNVP-style normalizing flow in PyTorch (fully spec-driven) behind a workload-agnostic interface; Prometheus/Pushgateway/cAdvisor + 4 provisioned Grafana dashboards provide platform observability, all orchestrated by Docker Compose with pytest suites in backend, scheduler, and worker.

ExperimentIQ — Query-Routing RAG over MLflow Experiments

FastAPI, MLflow, Qdrant, sentence-transformers

Code: github.com/HiteshRasineni/ExperimentIQ

- Built a standalone RAG assistant that answers natural-language questions over MLflow runs, flow configs, logs, and the dataset registry: a rule-based router sends structured questions (metrics, params, comparisons) to the MLflow Tracking search API and unstructured questions (logs, config text, notes) to Qdrant dense vector retrieval, with an LLM answer builder citing run IDs, metric values, and sources plus confidence scores.
- Shipped as a containerized FastAPI service (Qdrant + API via docker-compose) with a hand-authored 10-question benchmark harness: 10/10 routing accuracy, 10/10 answer accuracy, 0/10 hallucination rate, with per-category breakdowns (metric_search/range/comparison, log_retrieval, config_qa) and seed/eval scripts for reproducibility.

RESEARCH EXPERIENCE

Dark Matter Mono-Z (Density Estimation)

Leptonic Mono-Z Dark Matter Search with Neural Spline Flows

H. Rasineni, B. Chebrolu. arXiv preprint, 2026 (submitted for journal publication). *Leptonic decay channel.* DOI: 10.48550/arXiv.2607.13771

Code: github.com/HiteshRasineni/Leptonic-Mono-z-CMS2015-DarkMatter-Search

- Trained five neural spline flows on 2.32 fb^{-1} of CMS Run 2015D data (μμ/ee channels, 37-dim features) and set 95% CL signal-strength limits via an SR+VR profile-likelihood fit: μ < 0.018 (scalar), 0.036 (vector), 0.050 (axial-vector).

Hadronic Mono-Z Dark Matter Sensitivity with Flow Matching

H. Rasineni, B. Chebrolu. arXiv preprint, 2026. *Hadronic decay channel.* DOI: 10.48550/arXiv.2609.02923

Code: github.com/HiteshRasineni/CMS2015DarkMatterSearch-HTMHT-

- Built the MadGraph5_aMC@NLO → Pythia 8 → Delphes pipeline (1.44M selected events) with a conditional flow-matching background model; projected expected significances of 2.89σ, 7.62σ, and 7.41σ, with extra-jet kinematics carrying 53–71% of the discriminating power.

Likelihood-Based Unsupervised Anomaly Detection in CMS Dijet Events

B. Chebrolu, H. Rasineni, P.A. Immadi (equal contribution). arXiv preprint, 2026. DOI: 10.48550/arXiv.2609.06686

Paper: the-unlikely-lab.github.io/the-unlikely-lab/research/dijet-anomaly

- Unsupervised, model-independent anomaly search on 378,278 CMS Run 2011B HT events (√s = 7 TeV): a neural spline flow learned the SM multijet background from jet-substructure and event-shape observables; the 99.9th-percentile low-likelihood tail was validated with mass-decorrelation, permutation-null, and stability tests (>88% overlap across seeds, rank correlation >0.95).

Improving Discovery-Significance Stability in Higgs Event Classification

J.J. Pujari, P.A. Immadi, H. Rasineni, T. Bikku, R.S. Puppala. *Discover Artificial Intelligence* (Springer Nature), 2026. DOI: 10.1007/s44163-026-01683-5

- Supervised contrastive pre-training with a parallel FT-Transformer + XGBoost ensemble reduced fold-to-fold variance in Approximate Median Significance (AMS = 3.74) versus focal-loss training, enabling more stable threshold selection.

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript / TypeScript (React, Next.js)

ML/Data: PyTorch, scikit-learn, OpenCV, Pandas, Matplotlib, MLflow, Qdrant, Density Estimation (Normalizing Flows, Likelihood Ratios), RAG

Web/Platform: Next.js, React, FastAPI, TypeScript, Tailwind CSS, MDX

Tools/Infra: Git, GitHub Actions (CI/CD), Docker, Kubernetes, PostgreSQL, Redis, MinIO, Prometheus, Grafana, Pandoc, LaTeX, uproot, Google Colab

CERTIFICATIONS

- **Microsoft Certified: Azure AI Fundamentals** (Credential)
- **AWS Academy Graduate — Cloud Foundations (Training Badge)** (Credential)
- **WorldQuant University, Applied AI Lab — Deep Learning for Computer Vision** (YOLOv8, MTCNN, Inception-ResNetV1) (Credential)
- **IBM Machine Learning Specialist (Professional V1)** (EDA, ensembles, transfer learning) (Credential)
- **NVIDIA DLI — Fundamentals of Deep Learning** (Credential)
- **NVIDIA DLI — Transformer-based NLP** (Credential)